



Luigi RUSSO

Insegnamento di CHIMICA GENERALE E INORGANICA

Corso di laurea in SCIENZE AGRARIE E FORESTALI

SSD: CHIM/03

CFU: 6,00

ORE PER UNITÀ DIDATTICA: 48,00

Periodo di Erogazione: Primo Semestre

Italiano

Lingua di insegnamento	ITALIANO
Contenuti	Modelli Atomici. Configurazione elettronica e proprietà periodiche degli elementi. Nomenclatura. Stechiometria. Concetti di mole, numero atomico, massa atomica e molecolare. I legami chimici. Teoria VSEPR. Teoria del legame di valenza. Orbitali molecolari. Forze intermolecolari. Stati di aggregazione della materia. Stato Gassoso: Equazione di stato dei gas ideali, Teoria cinetica dei gas, Gas reali. Stato Liquido: proprietà generali. Stato Solido: tipi di solidi e loro proprietà. Soluzioni ideali e non ideali. Proprietà colligative. Equilibrio chimico. Principio di Le Chatelier. Costante di equilibrio. Proprietà degli acidi e delle basi. Equilibri acido-base. Curve di titolazione. Equilibri di solubilità. Principi della termodinamica. Velocità di reazione. Meccanismi.
Testi di riferimento	Nivaldo J. Tro - CHIMICA un approccio molecolare, Edises, Ed. II/2018 Bertini, Luchinat, Mani STECHIOMETRIA, Casa Editrice Ambrosiana, 2009.
Obiettivi formativi	Il corso di Chimica generale ed inorganica intende fornire le nozioni generali e la conoscenza dei principi fondamentali della chimica generale ed inorganica volti alla comprensione, a livello molecolare, dei processi chimici di interesse agrario e forestale. Al termine del corso, lo studente avrà acquisito conoscenze sulla struttura atomica e sulla tavola periodica, sui legami chimici e sugli equilibri in fase gassosa e acquosa. Nello specifico lo studente sarà in grado di: -riconoscere le principali classi di composti inorganici. - descrivere le proprietà microscopiche con particolare riferimento alle configurazioni elettroniche ed ai legami chimici. -scrivere e bilanciare le equazioni di reazione di acidi, basi, sali e redox. -ragionare criticamente sull'acidità o basicità delle soluzioni avendo ben chiaro il concetto di pH. -di eseguire le operazioni di base di un laboratorio di chimica inorganica (preparazione di soluzioni e utilizzo di semplice strumentazione, quale pH-metri, burette, etc.) e di applicare alcune delle più comuni tecniche di analisi volumetrica (titolazioni).
Prerequisiti	Il corso non prevede propedeuticità formali.
Metodologie didattiche	- Lezioni frontali in aula su tutti gli argomenti del corso. Le lezioni sono svolte mediante l'ausilio della lavagna e mediante la proiezione di diapositive - Esercitazioni numeriche in aula per la soluzione guidata di esercizi numerici mediante l'ausilio della lavagna - Esercitazioni di laboratorio. Le esercitazioni, della durata di circa 2 ore, riguardano gli argomenti trattati durante le lezioni frontali con lo scopo di mettere in pratica alcuni dei concetti di base sviluppati. Le attività di laboratorio richiedono frequenza obbligatoria e ogni studente è tenuto a produrre un proprio "quaderno di laboratorio" in cui sono descritte e discusse le esperienze svolte.
Metodi di valutazione	L'esame consiste nel superamento, con una votazione di almeno 14/30, di una prova scritta, della durata di 120 minuti, durante la quale lo studente, attraverso risoluzione di esercizi di stechiometria, dovrà applicare le conoscenze acquisite sulla nomenclatura, sulla reattività e sulla geometria molecolare dei composti inorganici. Il superamento della prova scritta è propedeutico all'esame orale. Durante lo svolgimento del corso verranno effettuate due valutazioni periodiche mediante prove scritte di verifica, della durata di 120 minuti ciascuna, con esercizi di stechiometria. Il superamento con 14/30 di entrambe le prove di accertamento esonererà lo studente dalla prova scritta. In carenza o insufficienza degli accertamenti periodici lo studente dovrà sostenere la prova scritta nelle date previste dal calendario di esami. L'esame orale è volto a valutare la capacità di ragionamento e di collegamento tra i vari argomenti del corso ed è costituito da domande sulla parte teorica del programma, in cui sarà valutata la capacità di organizzare l'esposizione, la padronanza della materia attraverso la capacità di espressione e di collegamento tra i diversi argomenti del programma. Saranno, inoltre, discussi gli aspetti pratici e teorici di una esercitazione di laboratorio. La valutazione finale sarà espressa in trentesimi e terrà conto dell'esito della prova orale (60%), della prova scritta (40%).
Altre informazioni	Allo studente è data la possibilità di sostenere due prove scritte di autovalutazione in itinere, strutturate in modo analogo all'esame scritto, ma limitate al programma svolto fino a quel momento, con successiva correzione. Potrà avvalersi della presenza di un esercitatore per ricevimento in aula in cui gli studenti potranno chiedere chiarimenti e proporre la risoluzione di esercizi. Lo studente potrà avvalersi del materiale didattico (dispense, esercizi, ecc.) messo a disposizione sul sito web del Dipartimento di Scienze e Tecnologie Ambientali Biologiche e Farmaceutiche (https://www.distabif.unicampania.it/dipartimento/docenti/) Il docente è disponibile per ricevimento studenti nei giorni indicati sulla scheda insegnamento e su richiesta inoltrata via e-mail.
Programma del corso	Proprietà e trasformazioni chimiche. Unità di misura. Errore di misurazione. Propagazione degli errori di misura. Cifre significative. Arrotondamento. Metodo dell'analisi dimensionale. Origini storiche ed evoluzione del concetto di struttura e composizione dell'atomo. Numero atomico e numero di massa. Isotopi. Molecole e ioni. Formula e nomenclatura di composti inorganici. Masse atomiche e molecolari. Concetto di mole e numero di Avogadro. Analisi chimica: composizione percentuale e formule empiriche. Principi di conservazione. Numeri di ossidazione. Classificazione delle reazioni chimiche e metodi per bilanciarle. Rapporti quantitativi tra le specie. Reagente limitante. Concetto di resa. Determinazione formula minima. I gas. Legge di Boyle, Charles-Gay Lussac, Avogadro, Dalton. Equazione di stato dei gas. Temperatura e pressione standard. Gas ideali. Teoria cinetica molecolare. Gas reali e deviazione dal comportamento ideale. Il modello quantomeccanico dell'atomo. Caratteristiche delle onde elettromagnetiche. Teoria dei quanti. Principio di indeterminazione di Heisenberg. Equazione di Schrödinger. Orbitali. Numeri quantici. Configurazione elettronica. Proprietà periodiche e Tavola periodica. Potenziale di ionizzazione. Affinità elettronica. Elettronegatività Legame chimico. Tipi di legame: covalente, dativo, ionico. Teoria di Lewis. Teoria VSEPR: geometria molecolare e polarità delle molecole. Ibridizzazione. Teoria del legame di valenza. Orbitali molecolari. Stati di aggregazione. Natura e classificazione delle interazioni intermolecolari. Legame a idrogeno. Le soluzioni. Rapporti relativi nelle soluzioni: molarità, normalità, frazione molare, molalità, percentuali. Fenomeni di solubilizzazione: insaturazione, saturazione e sovrasaturazione. Proprietà colligative e loro leggi. Equilibrio chimico. Natura e caratteristiche dello stato di equilibrio. Principio di Le Chatelier. Espressione della costante di equilibrio. Quoziente di reazione. Equilibri in sistemi omogenei e eterogenei. Metodi di calcolo per le concentrazioni all'equilibrio. Proprietà degli acidi e delle basi, Equilibri acido-base, Equilibri di solubilità. Definizioni di acidi e basi: Arrhenius, Bronsted e Lowry, Lewis. Specie anfiprotiche. Prodotto di autoionizzazione dell'acqua. Definizione di pH. Acidi e basi forti. Acidi e basi deboli. Acidi e basi poliprotici. Idrolisi. Soluzioni tampone. Curve di titolazione acido-base. Indicatori. Reazioni di precipitazione. Prodotto di solubilità. Effetto dello ione a comune. Principi della termodinamica. Aspetti energetici nelle trasformazioni: calore, capacità termica, entalpia, entropia, energia libera, fattori che influenzano la velocità delle trasformazioni chimiche, energia di attivazione. Il laboratorio di chimica: norme di sicurezza in laboratorio. Esercitazioni pratiche di laboratorio: 1) Equilibrio chimico. Misura del pH 2) Titolazione acido forte/base forte. Curva di titolazione

English

Teaching language	Italian
Contents	Atomic models. Electronic configuration and periodic properties of the elements. Nomenclature. Stoichiometry. Mole, atomic number, atomic and molecular mass. Chemical bonds. VSEPR theory. Valence bond theory. Molecular orbitals. Intermolecular interactions. States of matter. Gaseous state: State equation for ideal gases, Kinetic gas theory, Real gases. Liquid state: general properties. Solid state: types of solids and properties. Ideal and non-ideal solutions. Colligative properties. Chemical equilibrium. Le Chatelier principle. Equilibrium constants. Solutions of acids and bases. Acid-base equilibrium. Titration curve. Saline solutions. Thermodynamics principles. Kinetic of a reaction. Reaction mechanisms.
Textbook and course materials	Nivaldo J. Tro - CHIMICA un approccio molecolare, Edises, Ed. II/2018. Bertini, Luchinat, Mani STECHIOMETRIA, Casa Editrice Ambrosiana, 2009.
Course objectives	The aim of this course is to provide the fundamental principles of general and inorganic chemistry for fully understanding, at molecular level, the chemical processes of agricultural and forestry interest. At the end of the course, the student will have acquired the knowledge on atomic models, periodic table, chemical bonds, chemical equilibrium in gaseous and liquid phase. In particular the student will be able to: -identify the classes of inorganic compounds -describe the macroscopic properties with particular interest on Electronic configurations and chemical bonds - write and balance reaction equations for acids, bases, salts and redox - critically discuss the acid-base properties of aqueous solution well knowing the meaning of pH -perform basic operations of the inorganic chemistry laboratory (to prepare solutions and to use simple instruments as pH-meters, burette) and to apply some of the most common volumetric analysis techniques as titration.
Prerequisites	No propaedeutic.
Teaching methods	- Frontal lectures on all course topics. The lectures are performed with the help of the blackboard and by the projection of slides - Stoichiometry exercises. Guided numerical exercises in the classroom by using the blackboard. - Laboratory practical experiences. The experiences, lasting about 2 hours, concern the topics covered during the lectures with the aim of putting into practice some of the basic concepts developed. Laboratory activities necessitate compulsory attendance and each student is required to produce his or her own "laboratory notebook" in which experiences are described and discussed.
Evaluation methods	The examination consists in the completion, with a vote of at least 14/30, of a written examination, lasting 120 minutes, in which the student have to solve stoichiometry exercises, using the knowledge of chemistry nomenclature, reactivity, and molecular geometry of inorganic compounds. A successful completion of the examinations is necessary to take the oral examination. During the course, two written periodic evaluations will be performed, lasting 120 minutes, containing stoichiometry exercises. A completion of these examinations, with a vote of 14/30, will allow the student to take the oral examinations. The oral examinations are aimed to evaluate the capability of reasoning and connecting the various arguments of the course and is constituted by questions about the theoretical part of the course to evaluate the knowledge of the studied subjects and the capability to organize the exposition and connection of the diverse arguments. The theoretical and practical aspects of the laboratory practices will be also discussed The final vote will be express in thirtieth and will take in account the outcomes of oral (60%) and written (40%) examinations.
Other information	The student is allowed to take two written self-assessment tests during the course, similar to the written examination, but focused to the arguments already treated. The student is also allowed to ask questions about theoretical arguments or exercises during appropriate excercitation sessions. The student will be provided with learning material available on the website of Department of Environmental, Biological and Pharmaceutical Sciences and Technologies (https://www.distabif.unicampania.it/dipartimento/docenti/). The professor is available to receive the students in the days indicated in their syllabus or after a student request via e-mail.
Course Syllabus	Chemical proprieties and transformations. Units of measurements. Different types of errors in measurement and measurement error calculation. Rules for significant figures. General rule for rounding. Dimensional analysis. Historical origins and evolution of the concept of atom (structure and composition). Atomic number and mass number. Isotopes. Chemical symbols. Molecules and ions. Ions in solution. Nomenclature of inorganic compounds. Atomic and molecular weight. Mole and Avogadro number. Principle of conservation. Oxidation numbers. Chemical reactions and balancing method. Limiting reagent and reaction yield. Chemical analysis: percent composition and empirical formula. Gaseous state. Laws of di Boyle, Charles and Gay-Lussac. Avogadro's and Dalton's laws. State equation for a gas. Absolute temperature. Gas pressure. Ideal gases. Kinetic gas theory. Real gases. Van der Waals equation for real gases Atomic structure: The quantum mechanical model. 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